

Computational learning theory

计算学习理论

Machine Learning (Bilingual)

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- Support vector machines have largely been motivated and analysed using a theoretical framework known as **computational learning theory**¹, also sometimes called **statistical learning theory**.
- It originates from the **probably approximately correct** (PAC), learning framework (Valiant 1984).

¹Some contents from

https://www.cs.cmu.edu/~lwehbe/10701_S19/files/Lecture_13.pdf



Approximately Correct (AC) the algorithm must learn a function that is approximately close to the true function.

- In practice, it quantified by an error parameter ϵ

Probable (P) the learning algorithm should, with high probability (quantified by a parameter δ), learn a function that is approximately correct.

The PAC model has significantly influenced the development of ML algorithms and the understanding of how and why they work, particularly in the field of supervised learning.



- What general laws constrain inductive learning?
- Want theory to relate
 - Number of training examples
 - Complexity of hypothesis space
 - Accuracy to which target function is approximated
 - Manner in which training examples are presented
 - Probability of successful learning
- See annual Conference on Computational Learning Theory

How many training examples suffice to learn target concept

1. If learner proposes instances as queries to teacher?
 - learner proposes x , teacher provides $f(x)$
2. If teacher (who knows $f(x)$) generates training examples?
 - teacher proposes sequence $\{(x^1, f(x^1)), \dots (x^n, f(x^n))\}$
3. If some random process (e.g., nature) generates instances, and teacher labels them?
 - instances drawn according to $P(X)$

Problem setting:

- Set of instances X
- Set of hypotheses $H = \{h : X \rightarrow \{0, 1\}\}$
- Set of possible target functions $C = \{c : X \rightarrow \{0, 1\}\}$
- Sequence of training instances drawn at random from $P(X)$ teacher provides noise-free label $c(x)$

Learner outputs a hypothesis $h \in H$ such that

$$h = \arg \min_{h \in H} \text{error}_{\text{train}}(h)$$

Example: Learning decision trees



Take $X = (X_1, \dots, X_n)$ s.t. $X_i \in \{0, 1\}$, $Y_i \in \{0, 1\}$

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2^n

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How many possible trees?

$|H| = 2^{2^n}$

How many training examples needed to find the right tree?

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Generalizing beyond training is impossible unless we add assumptions
training examples are provided according to distribution $P(X)$



The true error of h is the probability that it will misclassify an example drawn at random from $P(X)$

$$\text{error}_{\text{true}}(h) \equiv \Pr[h(x) \neq c(x)]$$



Training error of hypothesis h with respect to target concept c

- How often $h(x) \neq c(x)$ over training instances D

$$\text{error}_{\text{train}}(h) \equiv \Pr_{x \in D}[h(x) \neq c(x)] = \frac{1}{|D|} \sum_{x \in D} \delta(h(x) \neq c(x))$$



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True error of hypothesis h with respect to c

- How often $h(x) \neq c(x)$ over training instances drawn at random from D

$$\text{error}_{\text{true}}(h) \equiv \Pr_{x \sim P(X)} [h(x) \neq c(x)]$$

Consider a hypothesis h and its

- Error rate over training data: $\text{error}_{\text{train}}(h)$
- True error rate over all data: $\text{error}_{\text{true}}(h)$

We say h **overfits** the training data if

$$\text{error}_{\text{true}}(h) > \text{error}_{\text{train}}(h)$$

Amount of overfitting =

$$\text{error}_{\text{true}}(h) - \text{error}_{\text{train}}(h)$$

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Can we bound $\text{error}_{\text{true}}(h)$ in terms of $\text{error}_{\text{train}}(h)$

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$$\text{error}_{\text{true}}(h) \equiv \Pr_{x \sim P(X)} [h(x) \neq c(x)]$$

$$\text{error}_{\text{train}}(h) \equiv \Pr_{x \in D} [h(x) \neq c(x)] = \frac{1}{|D|} \sum_{x \in D} \delta(h(x) \neq c(x))$$

$$\text{error}_{\text{true}}(h) \equiv \Pr_{x \sim P(X)} [h(x) \neq c(x)]$$

if D was a set of examples drawn from and independent of h , then we could use standard statistical confidence intervals to determine that with 95% probability, $\text{error}_{\text{true}}(h)$ lies in the interval:

$$\text{error}_D(h) \pm 1.96 \sqrt{\frac{\text{error}_D(h)[1 - \text{error}_D(h)]}{n}}$$

but D is the training data for h .



A hypothesis h is **consistent** with a set of training examples D of target concept c if and only if $h(x) = c(x)$ for each training example $(x, c(x))$ in D .

$$\text{consistent}(h, D) \equiv h(x) = c(x), \quad \forall (x, c(x)) \in D$$



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Version Space The **version space**, $VS_{H,D}$ with respect to hypothesis space H and training examples D , is the subset of hypotheses from H consistent with all training examples in D .

$$VS_{H,D} \equiv \{h \in H \mid \text{consistent}(h, D)\}$$



ϵ -**Exhausted** The version space $VS_{H,D}$ with respect to training data D is said to be ϵ -**exhausted** if every hypothesis h in $VS_{H,D}$ has true error less than ϵ .

$$\text{error}_{\text{true}}(h) < \epsilon \quad \forall h \in VS_{H,D}$$



How many examples will ϵ -exhaust the version space?

Theorem (Haussler, 1988) If the hypothesis space H is finite, and D is a sequence of $m \geq 1$ independent random examples of some target concept c , then for any $0 \leq \epsilon \leq 1$, the probability that the version space with respect to H and D is not ϵ -exhausted (with respect to c) is less than

$$|H|e^{-\epsilon m}$$



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Interesting! This bounds the probability that **any consistent learner** will output a hypothesis h with $\text{error}(h) \geq \epsilon$.

- Any(!) learner that outputs a hypothesis consistent with all training examples (i.e., an h contained in $VS_{H,D}$)

after m training examples is at most $|H|e^{-\epsilon m}$

$$\underbrace{Pr \left[\exists h \in H \text{ s.t. } (\text{error}_{\text{train}}(h) = 0) \wedge (\text{error}_{\text{true}}(h) > \epsilon) \right]}_{\delta} \leq |H|e^{-\epsilon m}$$

Suppose we want this probability to be at most δ

1. How many training examples suffice?

$$m \geq \frac{1}{\epsilon} (\ln |H| + \ln(1/\delta))$$

2. If $\text{error}_{\text{train}}(h) = 0$ then with probability at least $(1 - \delta)$:

$$\text{error}_{\text{true}}(h) \leq \frac{1}{m} (\ln |H| + \ln(1/\delta))$$



Example: H is Conjunction of up to N Boolean Literals

Consider classification problem $f : X \rightarrow Y$:

- instances: $X = (X_1, X_2, X_3, X_4)$ where each X_i is boolean
- Each hypothesis in H is a rule of the form:
 - IF $(X_1, X_2, X_3, X_4) = (0, ?, 1, ?)$, THEN $Y = 1$, ELSE $Y = 0$
 - i.e., rules constrain any subset of the X_i

How many training examples m suffice to assure that with probability at least 0.99, any consistent learner using H will output a hypothesis with true error at most 0.05?



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How many training examples m suffice to assure that with probability at least 0.99, any consistent learner using H will output a hypothesis with true error at most 0.05?

$$|H| = 3^4$$

$$m \geq \frac{1}{0.05} \left(\ln |H| + \ln \frac{1}{0.01} \right)$$

$$m \geq \frac{1}{\epsilon} (\ln |H| + \ln(1/\delta))$$



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Consider classification problem $f : X \rightarrow Y$:

- instances: $X = (X_1, \dots, X_N)$ where each X_i is boolean
- learned hypotheses are decision trees of depth 2, using only two variables

Example: Depth 2 Decision Trees



Consider classification problem $f : X \rightarrow Y$: $m \geq \frac{1}{\epsilon} (\ln |H| + \ln(1/\delta))$

- instances: $X = (X_1, \dots, X_N)$ where each X_i is boolean
- learned hypotheses are decision trees of depth 2, using only two variables

$$\binom{2}{N} \text{Trees} = \frac{N!}{(N-2)!2!} = \frac{N(N-1)}{2}$$

2^4 ways to label the nodes

$$|H| = 8N(N-1)$$

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How many training examples m suffice to assure that with probability at least 0.99, **any** learner that outputs a consistent depth 2 decision tree will have true error at most 0.05?

$$m \geq \frac{1}{0.05} \left(\ln(8N(N-1)) + \ln\left(\frac{1}{0.01}\right) \right)$$

Probably Approximately Correct (PAC) Learning

Consider a class C of possible target concepts defined over a set of instances X of length n , and a learner L using hypothesis space H .

Definition: C is **PAC-learnable** by L using H if for all $c \in C$, distributions D over X , ϵ such that $0 < \epsilon < 1/2$, and δ such that $0 < \delta < 1/2$, learner L will with probability at least $(1 - \delta)$ output a hypothesis $h \in H$ such that $\text{error}_D \leq \epsilon$, in **time that is polynomial** in $1/\epsilon$, $1/\delta$, n and $\text{size}(c)$.

Probably Approximately Correct (PAC) Learning

Consider a class C of possible target concepts defined over a set of instances X of length n , and a learner L using hypothesis space H .

Definition: C is **PAC-learnable** by L using H if for all $c \in C$, distributions D over X , ϵ such that $0 < \epsilon < 1/2$, and δ such that $0 < \delta < 1/2$, learner L will with probability at least $(1 - \delta)$ output a hypothesis $h \in H$ such that $\text{error}_D \leq \epsilon$, in **time that is polynomial** in $1/\epsilon$, $1/\delta$, n and $\text{size}(c)$.

Sufficient condition: Holds if learner L requires only a polynomial number of training examples, and processing per example is polynomial.



So far, assumed $c \in H$

Agnostic learning setting: don't assume $c \in H$

- What do we want then?
 - The hypothesis h that makes fewest errors on training data
- What is sample complexity in this case?

$$m \geq \frac{1}{2\epsilon^2} (\ln |H| + \ln(1/\delta))$$

Here ϵ is the difference between the training error and true error of the output hypothesis (the one with lowest training error)



- Given m independent flips of a coin with true $Pr(\text{heads}) = \theta$ we can bound the error in the maximum likelihood estimate $\hat{\theta}$

$$Pr[\theta > \hat{\theta} + \epsilon] \leq e^{-2m\epsilon^2}$$

- Relevance to agnostic learning: for any **single** hypothesis h

$$Pr[\underbrace{\text{error}_{\text{true}}(h)}_{\theta} > \underbrace{\text{error}_{\text{train}}(h)}_{\hat{\theta}} + \epsilon] \leq e^{-2m\epsilon^2}$$

- But we must consider all hypotheses in H

$$Pr[\exists h \in H | \text{error}_{\text{true}}(h) > \text{error}_{\text{train}}(h) + \epsilon] \leq |H|e^{-2m\epsilon^2}$$

- So, with probability at least $(1 - \delta)$ every h satisfies

$$\text{error}_{\text{true}}(h) \leq \text{error}_{\text{train}}(h) + \sqrt{\frac{\ln |H| + \ln \frac{1}{\delta}}{2m}}$$



- When estimating parameter θ inside $[a, b]$ from m examples

$$Pr(|\hat{\theta} - E[\hat{\theta}]| > \epsilon) \leq 2 \exp\left(\frac{-2m\epsilon^2}{(b-a)^2}\right)$$

- When estimating a probability θ is inside $[0, 1]$, so

$$Pr(|\hat{\theta} - E[\hat{\theta}]| > \epsilon) \leq 2 \exp(-2m\epsilon^2)$$

- And if we're interested in only one-sided error, then

$$Pr(E[\hat{\theta}] - \hat{\theta} > \epsilon) \leq \exp(-2m\epsilon^2)$$



$$m \geq \frac{1}{2\epsilon^2} (\ln |H| + \ln(1/\delta))$$

Here ϵ is the difference between the training error and true error of the output hypothesis (this holds for all h in H)

But, the output h with lowest training error might not give us the h^* with lowest true error. How far can true error of h be from h^* ?



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$$\text{error}_{\text{true}}(h) \leq \text{error}_{\text{true}}(h^*) + 2\epsilon$$

- h best training error hypothesis
- h^* best true error hypothesis



$$m \geq \frac{1}{2\epsilon^2} (\ln |H| + \ln(1/\delta))$$

Question: For $H = \{h|h : X \rightarrow Y\}$, if $|H|$ is infinite, what measure of complexity should we use in place of $|H|$?



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Question: For $H = \{h|h : X \rightarrow Y\}$, if $|H|$ is infinite, what measure of complexity should we use in place of $|H|$?

Answer: The largest subset of X for which H can **guarantee** zero training error (regardless of how it is labeled)



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VC dimension of H is the size of this subset.

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Informal intuition:

- decision tree example: how many labels do we need to see to learn h ?



Dichotomy a **dichotomy** of a set S is a partition of S into two disjoint subsets.

Dichotomy a labeling of each member of S as positive or negative.

Shattered a set of instances S is **shattered** by hypothesis space H if and only if for every dichotomy of S there exists some hypothesis in H consistent with this dichotomy.

Vapnik-Chervonenkis dimension the **Vapnik-Chervonenkis dimension**, $VC(H)$, of hypothesis space H defined over instance space X is the size of the largest finite subset of X shattered by H . If arbitrarily large finite sets of X can be shattered by H , then $VC(H) \equiv \infty$.

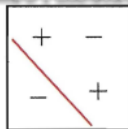
$$VC(\mathcal{H}) = \max\{m : \Pi_{\mathcal{H}}(m) = 2^m\}$$



存在这样的集合，其 $2^3 = 8$ 种对分均可被线性划分实现

(a) 示例集大小为 3

假设空间不能实现所有的对分



对任何集合，其 $2^4 = 16$ 种对分中至少有一种不能被线性划分实现

(b) 示例集大小为 4



How many randomly drawn examples suffice to ϵ -exhaust $VS_{H,D}$ with probability at least $(1 - \delta)$?

ie., to guarantee that any hypothesis that perfectly fits the training data is probably $(1 - \delta)$ approximately (ϵ) correct

$$m \geq \frac{1}{\epsilon} (4 \log_2(2/\delta) + 8VC(H) \log_2(13/\epsilon))$$

Compare to our earlier results based on $|H|$:

$$m \geq \frac{1}{\epsilon} (\ln(1/\delta) + \ln |H|)$$